

Integrating Machine Learning (ML) and Deep Learning (DL) Methods to Produce Reliable Wind Power Predictions

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Abstract— Wind energy forecasting is crucial for effective implementation due to weather instability and unpredictability. Seamless incorporation into contemporary electricity systems. Various methodologies and strategies have been used and utilized to achieve precise wind power prediction over a wide range of timeframes. Every approach has advantages and disadvantages. This study first examines the current uses of Machine Learning and Deep Learning methods in wind power and wind speed predictions. An extensive examination and classification of the most significant forecasting techniques has been conducted. Simultaneously, thought-provoking conversations are offered to facilitate the examination of all the topics presented. Two algorithms, one from machine learning (ML) and the other from deep learning (DL), have been chosen and used to anticipate wind power. The findings have shown the strong capacity of both algorithms to accurately capture the characteristics of the produced power. Additional analyses of the outcomes about their correctness and prospects are being contemplated.

Keywords— Machine Learning, Deep Learning, Wind Power Predictions, Wind Speed, Forecasting Approaches.

I. INTRODUCTION

Renewable sources of energy are extensively used to encounter the increasing energy demand and to contribute towards the objective of achieving carbon neutrality.[1] Some of these objectives include reducing greenhouse gas emissions, transitioning to renewable energy sources, and ensuring affordable and accessible energy services for all communities. One of the most crucial renewable energy sources for achieving the aforementioned goals is wind energy.[2] Over the last twenty years, the capacity of wind power has seen substantial growth due to increased profitability and reliability in the wind sector. When considering all other renewable energy sources, wind power also generates the least amount of carbon dioxide. [3] To achieve the energy transition and guarantee energy security on all scales—from local to global—wind power is essential. However, the generation of wind power is sporadic and heavily dependent on unpredictable weather. Consequently, integrating significant quantities of wind power into the electrical system may give rise to several technological

difficulties in managing and controlling the distribution and regulation of electricity. Precise power production forecasting is essential for affordable and steady pricing in today's energy markets, where variable renewable energy sources (RESs) like solar and wind play a major role.[4] However, predicting wind power has always been a difficult task, which leads to increased uncertainty about the management and control procedures, as well as the integration strategies utilised in the market for energy. Unpredictable weather, restrictions imposed by the electrical system, and possible turbine faults are the main factors influencing wind power predictions. Advanced machine learning and deep learning methods are used to assist those making decisions to optimise power generation, delivery, and energy trading operations, as well as to increase forecasting accuracy. [5-7] This work provides an extensive overview of research publications that explore the use of certain machine learning (ML) and deep learning (DL) methods for wind energy prediction. Furthermore, a case study is provided where wind measurements were utilised to predict wind power using ML and DL algorithms. [8-9] Support vector regression, a subset of support vector machines, techniques used for regression and approximation, is the recommended machine learning approach. Long-short memory, a member of the recurrent neural network (RNN) family, is the deep learning technique that is used. These two methods are used in this research to anticipate wind power based on data gathered on-site. A preliminary study of the projected outcomes of both algorithms is presented. [10] However, the primary goal of the study is to set the stage for a deeper investigation where forecasting of wind power will be carried out utilising a hybrid approach that incorporates ML and DL algorithms, as opposed to comparing the outcomes.

The primary benefits of this enquiry are as follows:

Recent research, offers a thorough analysis of deep learning and machine learning to power forecasting and wind speed.

A critical analysis of these approach usage in wind energy forecasting is presented, along with a discussion of recent developments and potential uses.

The results evaluated the wind power, using specific deep learning and machine learning techniques that are presented and analysed in the last part.

The following sections make up the remaining half of the article. The machine learning and deep learning techniques used in this study to forecast wind speed are described in Section II. Section III showcases and discusses a few chosen outcomes. Sections IV and V conclude the work and outline the next tasks respectively.

II. METHODOLOGY

The current study utilises wind measurements to use a Deep Learning method (Long Short-term Memory) and a method from Machine Learning (Support Vector Regression) to predict wind power for a certain hour. The subsequent sections will include a mathematical explanation of each method.

A. SVR-ML algorithm

The SVR ML uses an algorithm for regression and estimation. One supervised machine learning technique that may be used for both regression and classification is the SVM. The following is the forecasting process that makes use of SVR. For a collection of measurements made up of an input-output pair dataset

$$\{(x_1, y_1), \dots, (x_n, y_n)\} \in R^d \times R,$$

where R^d is the amount of the training data, space containing the input structures x_i with dimensions d , and the goal variables n and $y_i \in R$.

The SVR model's regression issue may be stated as follows (1):

$$f(x) = w^T \cdot x + b \quad (1)$$

In this case, $(\cdot)$ denotes the dot product, w is the weight vector, b is the bias component, and f is the function to estimate the outputs R^d .

One way to summarise the SVR optimisation issue is to solve the equation above to determine w , based on the minimisation of (2):

$$\min \left\{ \frac{1}{2} w^2 + C \sum_{i=1}^n (\xi_i + \check{\xi}_i) \right\} \quad (2)$$

With constraints in equation (3) to (5)

$$y_i - w^T \cdot \theta - b \leq \varepsilon + \xi_i \quad (3)$$

$$-y_i + w^T \cdot \theta - b \leq \varepsilon + \check{\xi}_i \quad (4)$$

$$\xi_i, \check{\xi}_i \geq 0 \quad (5)$$

Whereas ξ_i & $\check{\xi}_i$ is used C is the penalty set of the fault that is used to confirm the trade-off between the regularity period and observed risk, hence constraining the reliability of the estimate function ε . The variables are used towards cope

with mistakes generated by outliers. Lagrange multipliers, α_i and α_i^* may be used to solve the SVR minimisation issue. Finally, one may deduce (6) satisfying equations (2) to (5) by integrating Lagrange variables, which results in the formulation that follows:

$$f(x) = \sum_{i=1}^n (\alpha - \alpha_i^*) K(x, x_i) + b \quad (6)$$

K indicates a kernel function that allows the input information to be converted into a space of higher dimensions. A linear issue in a higher measurement space might then be produced by the nonlinear regression formula in the input space. This linear function, the polynomial, the radial base function, and the exponential radial basic function are some examples of kernel functions. Users may additionally modify the kernel function based on the needs of their application. The forecasting results for both systems are validated using the R^2 error, which is computed as shown in (7).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (7)$$

Where $\hat{y}_i = f(x_i)$ are the anticipated values, and the output variables $\bar{y}_i$ mean value is y_i .

B. LSTM-DL algorithm

A specific kind of recurrent neural network, also called an LSTM is made up of three thresholds: input, output, and forgetting gates. The vanishing error issue in RNNs was the motivation for the development of the gradient-based technique known as LSTM. The architecture of one unit inside the network known as LSTM is shown in Figure 1. One may get a thorough explanation of how an LSTM network operates. By defining the threshold, the forgetting gate—which stands for reservation and forgetting—defines the rise and decrease of the data flow. Because the hidden layer of an RNN only has one state, there is a serious problem with gradient explosion and disappearance. In contrast to RNN, LSTM appends the cell state structure, which highlights the LSTM network's large memory capacity and can maintain the cell state for a long time. The LSTM network has the potential to significantly improve wind power forecasting accuracy when working with massive volumes of wind power and wind speed data.

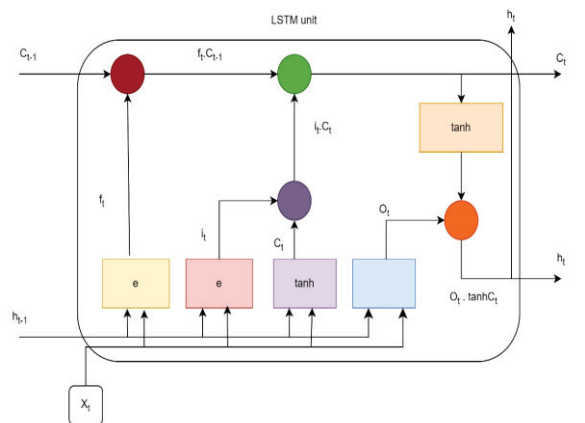


Fig. 1. An illustration of an LSTM network unit's internal structure.

At first, the neural network that makes up the LSTM decides which cell state information to retain and which to disregard. Equation (7) provides the corresponding function in exchange for the unit state information.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (7)$$

where σ stands for the sigmoid function, W_f for the forgotten gate's weight matrix, and b_f for the forgotten gate's bias term. Two variables, i_t and $\tilde{C}_t$, which are related to the prior moment as stated in equations (8) and (9), were generated by the input gate.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8)$$

$$\tilde{C}_t = \tanh(W_t \cdot [h_{t-1}, x_t] + b_c) \quad (9)$$

Cell state C_t , which, both before and after LSTM, is the transmission core of the cell state and has the following function (10).

$$C_t = C_{t-1} \circ f_t + i_t \circ \tilde{C}_t \quad (10)$$

Two parts contribute to the output gate h_t . The current cell state is the source of the other component, which may be stated in equations (11) and (12). The input of the present instant is derived from the output of the previous moment.

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (11)$$

$$h_t = O_t \circ \tanh(C_t) \quad (12)$$

Where $\circ$ is the multiplication by elements.

The flowchart for both methods, which illustrates the wind power forecasting procedure, is shown in Figure 2. The data is divided into two pieces once it has been scaled. A portion for testing and another for instruction. Of the whole dataset, 81% is the one used for the training dataset set and just 21% is the test dataset set. Subsequently, the accuracy parameter and hyperparameters are optimised by evaluating and training the model. The model is trained using the training dataset to assess its correctness and performance. The test dataset is then used to calculate the R^2 -error parameter. Upon obtaining the ideal parameters, wind power output projections are ultimately carried out.

III. DISCUSSIONS AND RESULTS

The machine learning and deep learning algorithms were trained using hourly wind speed and wind energy generation measurements collected over a year at a particular city location. The wind speed readings gathered throughout the year from the online Renewables database were used to calculate the turbine output statistics. A ninja standing eighty meters above the ground. With a 2MW rated capacity, the Vestas V90/2000 turbine type is selected as a typical wind turbine. The speed of the wind was measured at a height of 80 meters, which is its hub height.

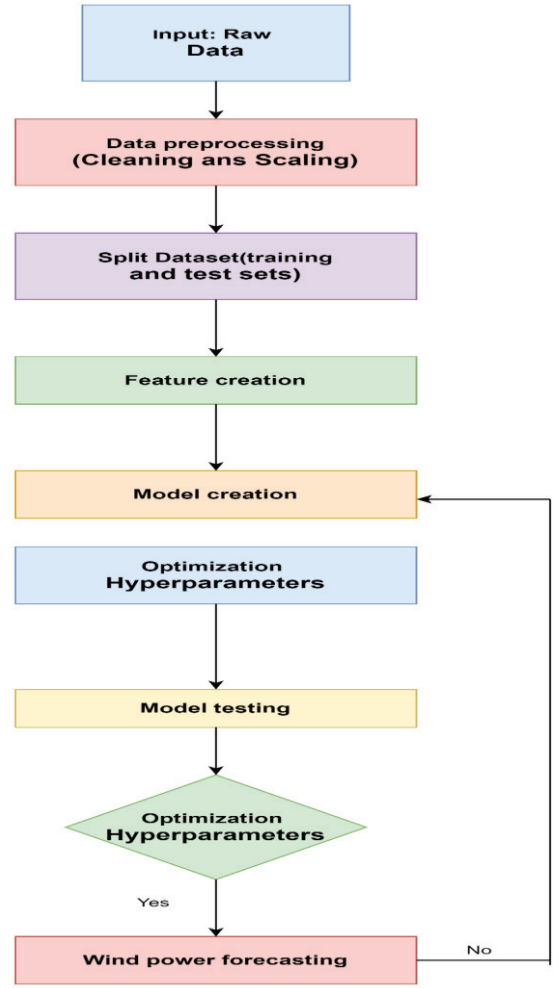


Fig. 2. The Two Algorithms' Flowcharts.

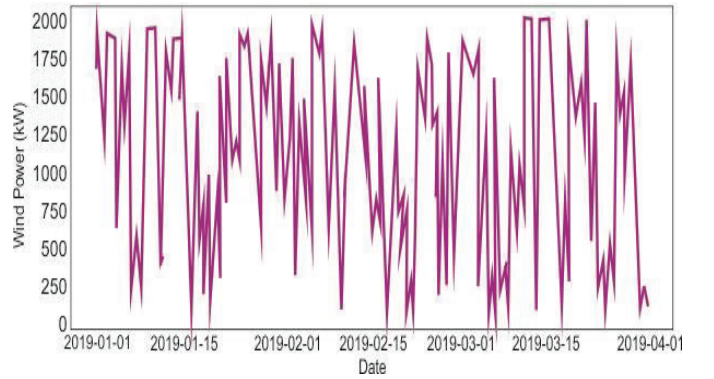


Fig. 3. One-year wind turbine power production.

The graphic in Figure 3 displays the wind turbine outputs over 3 months, using hourly observations. The conditions of the wind in the studied location have a significant impact on the turbine's power generation, which is intermittent. Periods of blackouts and unstable power generation may last anywhere from a few hrs up to several days. For the SVR model, the RBF kernel has been selected as the best option because of its remarkable capacity to precisely characterise the non-linear behaviour of wind measurements. Three hidden layers, each with thirty LSTM memory units, were used to build an LSTM model. Every layer has a 0.20 dropout rate, and the final dense layer predicts a single value. The wind energy measurement graph and the results of the

Long Short-Term Memory (LSTM) and Support Vector Regression predictions are shown in Figure 4. Both models demonstrated exceptional proficiency in estimating wind power for future days. The R^2 error is calculated to be 0.98 for SVR and 0.95 for LSTM. The SVR demonstrated superior performance compared to the LSTM in capturing the wind turbine output profile.

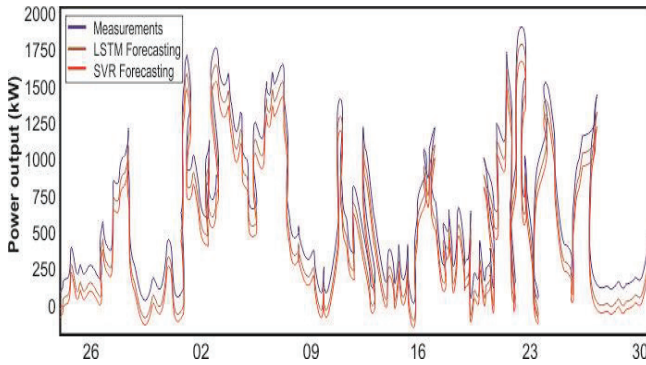


Fig. 4. Measurement and prediction of the power output of the wind turbine.

Figure 5 presents scatter plots of the power output measurements versus the predictions of the two algorithms, to provide a clearer illustration of their performances. It is evident that both plots accurately represented the trend of the measurement curve, albeit there was somewhat greater variation in the plot corresponding to the LSTM prediction.

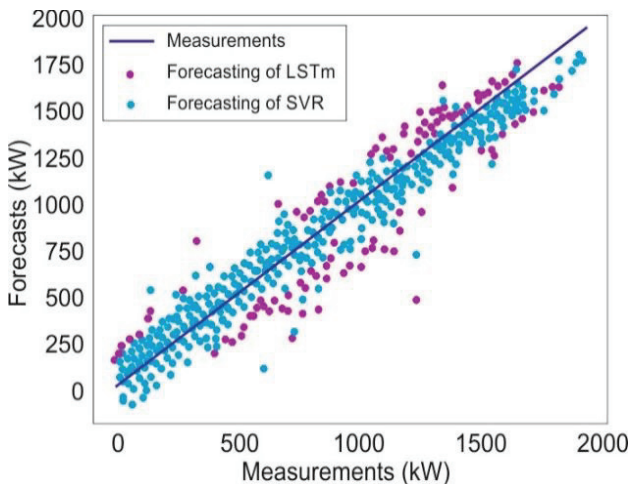


Fig. 5. LSTM and SVR-based scatter graph of the observed and predicted wind power output.

IV. CONCLUSION AND FUTURE WORK

This study offers a cutting-edge review of recent works concerning the usage of ML and DL for wind speed and power prediction. Because there are so many papers regarding AI-enabled algorithms utilised for forecasting in the wind power business, it was difficult to create a complete list of all the techniques and methodologies. The two algorithms: are SVR, a DL method, and LSTM, an ML algorithm. The findings of forecasting wind power based on hourly wind turbine output based on the wind data show that the SVR performs better than the LSTM in terms of wind power accuracy. This conclusion is supported by many additional studies and contradicted by other research. It suggests that further investigation is needed to assess different AI-enabled techniques. The comparison should involve testing the performances of the different methods

under a range of limitations, including dataset size, computing cost, and the effect of hyperparameters on prediction accuracy using different accuracy metrics. However, the main objective of this study is to pave the way for future research that will use hybrid techniques that mix ML and DL algorithms, rather than comparing the accuracy of wind power forecasts between the two systems.

A. Future work:

- An extensive state-of-the-art review covering further ML and DL algorithms, together with evaluations of the results that have previously been published, their shortcomings, and potential future paths.
- Comparing the forecasting capabilities of many algorithms based on a variety of variables, including how the complexity and computation time affect the predicting accuracy using various datasets,
- The development and use of hybrid methods based on the most accurate ML and DL algorithms selected.

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